

Caleb Shin

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EDUCATION

University of California, Santa Cruz

Anticipated Graduation: June 2028

Bachelor of Science, Electrical Engineering

Awards and Honors: Dean's List (Fall 2024)

Leadership Involvement: Formula SAE Electric, College Nine College Student Life

EXPERIENCE

Vehicle Dynamics Lead, Formula Slug FSAE EV

June 2025 - Present

- Serve as the primary vehicle dynamics resource for 200-member organization
- Research and govern mechanical design decisions across the full car
- Partner with the data acquisition team to analyze real-world vehicle behavior and build sensors
- Evaluate and optimize vehicle design through CAD modeling and FEA simulations
- Advocate for crucial designs within cost, manufacturing, and integration constraints
- Deliver lectures on physics fundamentals and applications in automotive engineering
- Founding member of aerodynamics team at Formula Slug

Composites Engineer, Formula Slug FSAE EV

September 2024 - Present

- Fabricate precision carbon fiber components via prepreg, wet layup, and infusion methods
- Perform stress and failure analysis on components, optimizing parts for relevant load patterns
- Design SAE-compliant assemblies emphasizing design for manufacturability and serviceability
- Operate and maintain vacuum systems, high-temperature curing ovens, and HVLP sprayers
- Manage and track inventory of volatile chemicals and highly valuable materials

Undergraduate Researcher, NECTAR Lab at UCSC

January 2026 - Present

- Model and analyze electrical, chemical, thermal, and optical behaviors of the light-assisted high-J electrolysis project in Ansys EDT (Maxwell/HFSS/Icepak), Fluent, and Speos
- Accelerated the lab's ability to iterate on designs through computer simulations
- Design and manufacture the gas collector, optical chamber fittings, and experiment setups
- Operate laboratory equipment during controlled trials and for post-experiment data collection
- Develop technical diagrams in support of manuscript preparation and internal communication

PROJECTS

Automotive Steering and Suspension Models

September 2025 - Present

- Programmed in Python using NumPy, SciPy, and Matplotlib, with supplementary work done in MATLAB/Simulink and Jupyter Notebooks
- Several multi-DOF physics models that simulate complex steady-state mechanical phenomena
- Built around efficient parametric workflows for rapid design iteration and scalability
- Validation through on-car telemetry/sensor system and real-world vehicle testing
- Analyzed models and research findings to identify performance gaps and optimize designs, accomplished +30% maximum lateral acceleration boost in our EV racecar (2.3G → 3.0G)

Mechanical Keyboard: PCB Design & Embedded Systems

June 2021 - August 2021

- Built a mechanical keyboard from a bare PCB kit, soldering 100+ through-hole components
- Firmware flashing and troubleshooting using ATmega328P and MCP23008 microcontrollers
- Modified off-the-shelf PCB in KiCad to redesign USB port footprint and customize silkscreen, then fabricated revised boards through PCBWay
- Gained fundamental circuitry experience with diodes, resistors, capacitors, fuses, and crystals
- Utilized DMM to find continuity faults and confirm operational voltages/currents of hardware

Custom Auto Body Panel Manufacturing

April 2025 - November 2025

- Independently designed and fabricated eight sheet metal body panels for UCSC's EV racecar
- Extensive operation of industrial fabrication equipment including press brake, foot shear, roller, turret punch, CNC router, and several small components and jigs made with 3D printers

SKILLS

Ansys (Fluent/Mechanical/Zemax OS/Speos/EDT/SpaceClaim), FEA, CFD, Python (NumPy/SciPy/Matplotlib), MATLAB/Simulink, C, C++, RISC-V Assembly, LaTeX, Onshape, 3D Printing, PCB Design, Embedded Systems, Sheet Metal Fabrication, Electrical Circuitry, Mechanical Engineering, WSL, GD&T, DFS, DFM